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The Influence of the Release Angle on the Damping Coefficient of a Simple Pendulum

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1. Research Design

1.1 Introduction

The nature of our world is intrinsically related to physics and how it governs physical space. To obtain even the smallest amounts of this knowledge, a wide range of experiments must ensue using empirical procedures and instruments; one such instrument is the Pendulum, a body of mass fixed at a particular point that can swing back and forth due to gravity (Britannica, 2024). This instrument is used everywhere from metronomes to park swings and even within seismographs. However, its main purpose is to depict Newtonian motions while modelling oscillatory behaviour, a study that allows scientists from across the world to investigate wave patterns, gravity, and simple harmonic motion. When the theory behind pendulums is taught, it is said that energy conversion will create an endless swing in accordance with Newton's first law of motion (Neal, 2019). However, due to external forces such as air resistance and energy dissipation, damping occurs. Countless scientists have attempted to deepen our understanding of damping, but a large area remains unexplored, especially regarding the damping coefficient. As a result, this scientific investigation aims to further pre-existing knowledge by investigating the effect that the release angle has on the damping coefficient.

1.2 Research Question

To what extent does an increase in the release angle of a pendulum (3°, 6°, 9°, 12°, 15°) influence its damping coefficient (Ns/m) in Simple Harmonic Motion when keeping the mass of the ball and length of the string constant?

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1.3 Background Information

Simple Harmonic Motion:

Simple harmonic motion is defined as a periodic motion in which an object oscillates along a straight line. The acceleration is towards a fixed point in that line and is proportional to the negative displacement from that point (SATHEE, 2024). This motion occurs in different systems, one within a spring mass system and the other along a simple pendulum.

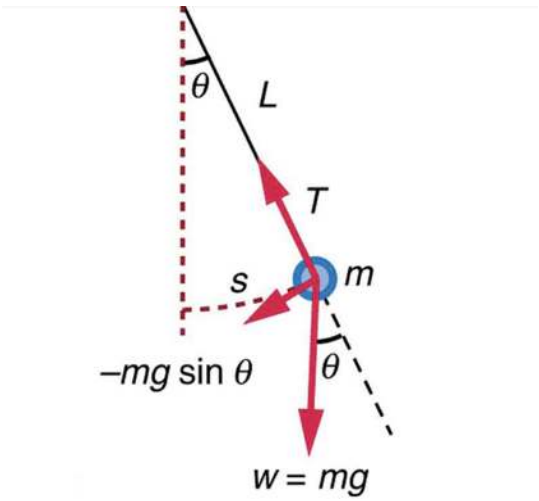


Figure 1: The free-body diagram that represents simple harmonic motion in a pendulum

(Douglas College)

Hooke's law states that an object performs simple harmonic motion when acted upon by a restoring force that is proportional to the displacement.

$$F = -kx$$

Where:

F - The restoring force (N)

k - The proportional constant or spring constant (N/m)

x - The displacement from the equilibrium position (m)

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However, since the simple pendulum system is being used Hooke's law is rewritten as demonstrated in Figure 1:

$$F = -mgsin\theta$$

Where:

F - The restoring force (N)

mg - The gravitational force (N)

θ - Angle/ Angular displacement (rad)

Within pendulums, when the mass is displaced from the equilibrium position (as long as the displacement is small), the restoring force approximately obeys Hooke's law. Beyond a certain displacement, or release angle, the pendulum will become an anharmonic oscillator (OpenStax, 2023). Due to this, the investigation will only consider smaller angles that allow the pendulum to obey Hooke's law.

Damping:

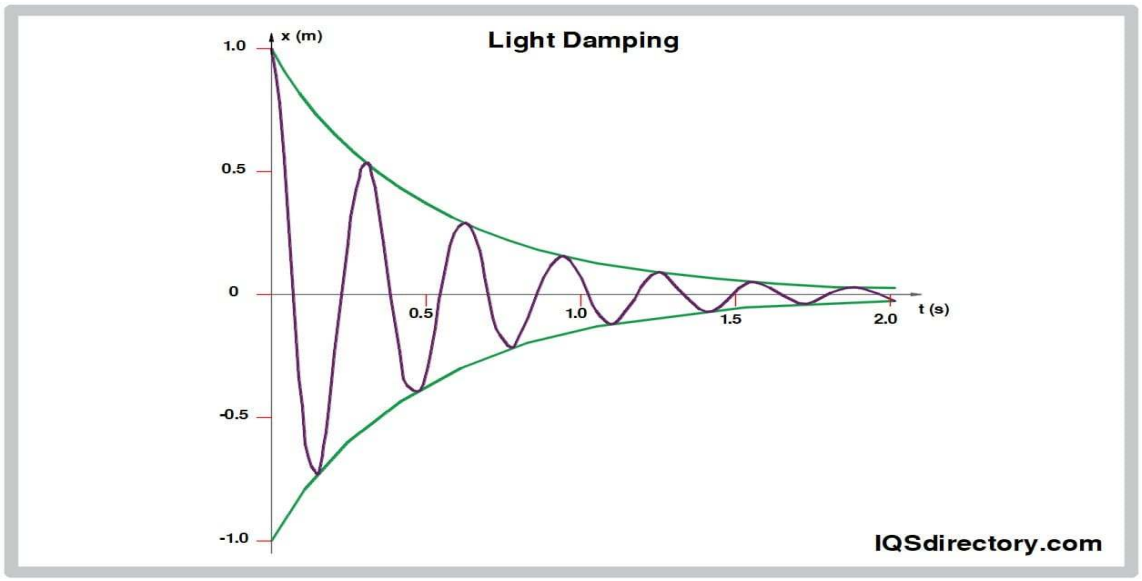


Figure 2: Represents an exponential decrease of amplitude as defined in an underdamped system

(IQSdirectory)

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Damping is defined as the reduction in oscillations and vibratory force within a particular system due to the dissipation of energy (Britannica, 2024). It is typically formed within a frictionless oscillating pendulum by the continuous motion of the mass. (Szyszkowski and Stilling, 2005). As mentioned previously, the restoring force is proportional to the displacement causing simple harmonic motion. However, when a frictional force proportional to the velocity is applied, the system will undergo damped harmonic motion. There are various scenarios in which damping exists; this investigation will focus on underdamping the process in which a pendulum swings and overshoots its rest position, causing a decrease in amplitude over time (element14, 2023).

The Damping Coefficient:

The damping coefficient quantifies the magnitude of damping in a system and is expressed as a function of velocity (Science Direct, 2019). As a result, the overall damping is directly influenced by the damping coefficient and the factors that affect it; similar to how a black hole is contingent on its singularity. In 2017, a study conducted by Pirooz Mohazzabi and Siva P. Shankar, titled Damping of a Simple Pendulum Due to Drag on Its String, derived a formula to correlate the amplitude to the damping coefficient.

$$A = A_0 e^{-kt/2} \quad \textbf{- (1)}$$

Where:

A - The amplitude measured at any point in time (m)

A_0 - The initial amplitude (m)

k - The damping coefficient (Ns/m)

t - Time (s)

After rearranging the formula:

$$ln(\frac{A}{A_0}) = -\frac{k}{2}t$$

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$$k = -2\frac{ln(\frac{A}{A_0})}{t} \quad \textbf{- (2)}$$

Hence, the graph of $ln(\frac{A}{A_0})$ versus t results in a straight line with a gradient $-\frac{k}{2}$ (Mohazzabi and Shankar, 2017).

The Release Angle:

The release angle is the angular position of the object at the start of the motion or when t=0. As aforementioned, the angle must remain small for the object to perform harmonic oscillations ($\theta \leq 15°$). This is because the restoring force is only proportional to the displacement at small angles.

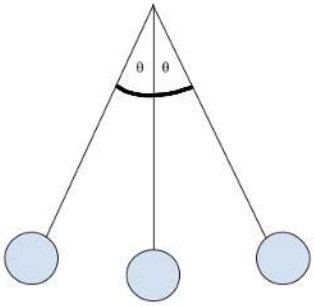


Figure 3: The motion of a simple pendulum during simple harmonic motion in conjecture to the release angle (Self-Drawn)

An understanding of trigonometry can be used to connect formula (2) with the release angle:

$$sin(\theta) = \frac{A_0}{L}$$

$$A_0 = Lsin(\theta) \quad \textbf{- (3)}$$

Where:

θ - The Release Angle (rad)

A_0 - Initial Amplitude (m)

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L- Length of the String (m)

The Final Derivation:

Plugging (3) into (2):

$$k = -2\frac{ln(\frac{A}{Lsin(\theta)})}{t} \quad \textbf{- (4)}$$

Above is a derived formula that will be utilised to calculate the damping coefficient for this investigation.

The time taken to damp will also be considered using scientific apparatus to ensure the validity of the derived formula, as an increase in the damping coefficient corresponds with an increase in the damping effect.

1.4 Hypothesis

The general understanding is that the damping effect increases with amplitude due to higher air resistance and energy dissipation (Tamura, 2012). **Hence, as the release angle increases, the damping coefficient will also increase due to the corresponding rise in amplitude.**

Null hypothesis: There is no significant relationship between the release angle and the damping coefficient of a simple pendulum.

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1.5 Variables

Table 1:

Manipulated and Measured Variables			
Variable		**Explanation**	**Apparatus**
Independent Variable	The Initial Release Angle	3°, 6°, 9°, 12°, 15° The variable will be altered in 3° increments to maintain harmonic motion while creating a variation to test the hypothesis.	Protractor
Dependant Variable	The Damping Coefficient	Measured by using the derived formula: $k = -2\frac{ln(\frac{A}{Lsin(\theta)})}{t}$ Experimentation will determine how time and amplitude vary with changes in the angle. Verification of the damping effect will be conducted using the Smart Gate.	Smart Gate

Table 2:

Controlled variables		
Variable	**Significance**	**Means of control**
Length of the String	The length of the string can affect the time period (increase or decrease) of the harmonic motion. Hence, it must be kept constant to ensure accurate and verifiable results.	Ensuring the length is fixed at (1.0000 ±0.0005m)
Mass of the Ball	The mass of the ball greatly affects the time period and drag coefficient. Hence, it must be kept constant to ensure accurate and verifiable results.	Ensuring the mass is fixed at (0.0500 ±0.0001kg)
The Time for the Amplitude at any given point	The amplitude at any given point is a variable in the derived formula. To ensure accurate and verifiable results, the point at which the amplitude is taken must be kept constant.	Ensuring that the amplitude is measured at time t=2s for all trials.
Experimental Set-up	The experimental setup affects all variables	Using the same

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	and can greatly skew the results. Hence, it must be kept constant to ensure accurate and verifiable results.	equipment during the data collection process

1.6 Materials and Set-up

1. Weight (0.0500 ±0.0001kg)
2. String (1.0000 ±0.0005m)
3. Apparatus stand
4. Pasco Smart Gate
5. Smart Gate Software (SPARKvue)
6. Laptop
7. Protractor (± 0.05°)

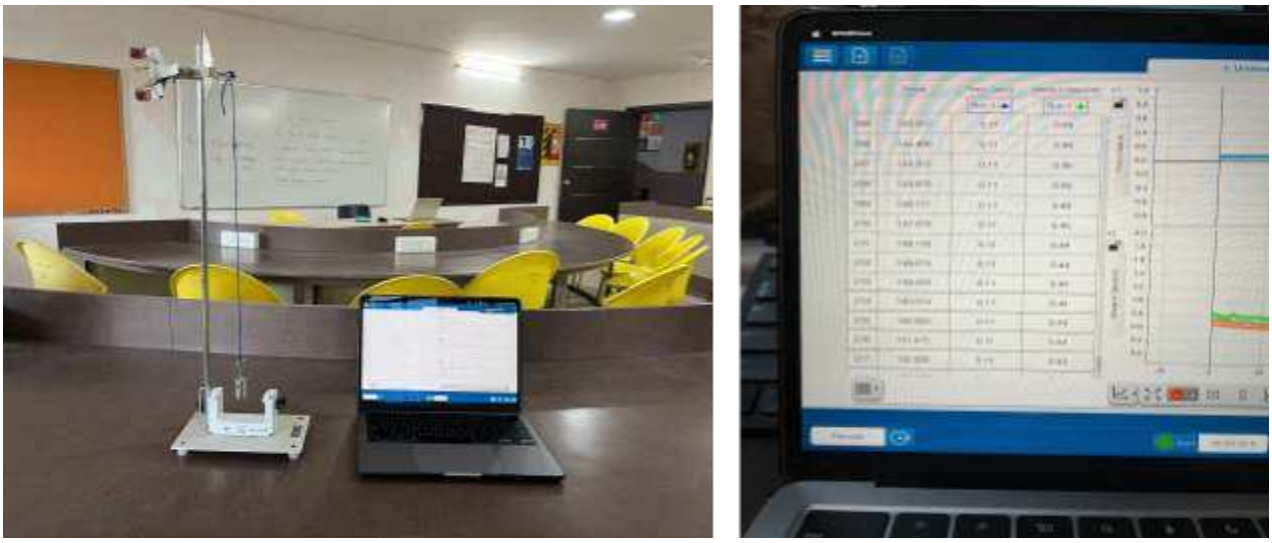


Figure 4: Experimental Set-Up

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1.7 Methodology

1. Install the Smart Gate software into a designated laptop.
2. Connect the Smart Gate to the laptop and then the apparatus stand.
3. Attach the ball of constant mass (0.05kg) to the string of length 1m.
4. Attach the ball-string system to the apparatus stand.
5. Start recording on the Smart Gate.
6. Position the ball at a displacement of 3° from the equilibrium.
7. Release the ball at this position.
8. Record the time and amplitude data with the software.
9. Repeat the experiment 5 times for this release angle.
10. Increase the degree by an increment of three, then repeat the process from step 5 to step 8 until a release angle of 15° is reached.
11. End all recordings.

1.8 Risk Assessment

Consideration type	Significance
Safety Concerns	The experiment presents no significant safety concerns, as it does not involve hazardous or toxic equipment. However, precautions should be taken to minimize contact with the mass during the procedure to mitigate any risk of injury.
Ethical Concerns	The experiment does not possess any ethical concerns as both humans and animals are not involved in the data collection process.
Environmental Concerns	The experiment does not pose harm to the environment as the entire procedure will not generate waste or cause degradation.

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1.9 Preliminary Trials

Before the final experiment, preliminary trials were conducted to identify room for error while optimising the experimentation process. In the initial stages, the photogate feature was used to record time and velocity. However, many data points were missing, which greatly hindered the data collection. After this experience, it was decided that the smart pulley feature would be used instead, as all three data sets were available and could be recorded accurately. This method also improved the accuracy of the data and its presentation.

2. Results:

2.1 Raw Data:

By using the SPARKvue software, data for the entire motion was recorded. Below are the data tables for the Total Run Time and the Amplitude.

Quantitative Data:

Table 3:

Angle (± 0.05) (deg)	***Total Time (s)***				
	Trial 1	**Trial 2**	**Trial 3**	**Trial 4**	**Trial 5**
3.00	162.300	160.780	160.317	161.318	161.738
6.00	166.370	165.689	165.590	166.113	165.120
9.00	167.991	169.235	170.362	168.561	168.234
12.00	171.026	172.002	171.803	171.563	171.997
15.00	174.101	174.967	176.166	175.316	175.234

Table 4:

Angle (± 0.05) (deg)	***Amplitude (m) (At t=2s)***				
	Trial 1	**Trial 2**	**Trial 3**	**Trial 4**	**Trial 5**
3.00	0.0500	0.0498	0.0513	0.0544	0.0499

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6.00	0.0568	0.0578	0.0612	0.0587	0.0621
9.00	0.0634	0.0612	0.0597	0.0634	0.0644
12.00	0.0653	0.0652	0.0679	0.0697	0.0726
15.00	0.0745	0.0758	0.0730	0.0745	0.0789

Based purely on the Quantitative data, the damping effect did not seem to increase with the release angle. This contradicted the original hypothesis, leaving room for further investigation.

Observed data (Qualitative with analysis):

After analysing the raw data from the Smart Gate, a summary of the analysis of the damping effect is provided, featuring excerpts from two of the trials. To maintain conciseness, the evidence has been limited to 25 data points across 3 trials, selected from a total dataset of over 200 data points from 5 trials.

Table 5:

3° Data Table (25 Data Points)									
No.	**Time (s) Run 1**	**Time in Gate (s) Run 1**	**Speed in Gate (m/s) Run 1**	**Time (s) Run 2**	**Time in Gate (s) Run 2**	**Speed in Gate (m/s) Run 2**	**Time (s) Run 3**	**Time in Gate (s) Run 3**	**Speed in Gate (m/s) Run 3**
1	0.345	0.06	0.80	0.387	0.08	0.75	0.412	0.07	0.78
2	0.789	0.06	0.80	0.897	0.08	0.73	0.989	0.07	0.78
3	1.567	0.06	0.79	1.156	0.08	0.73	1.656	0.07	0.78
4	2.123	0.06	0.78	2.735	0.08	0.73	2.525	0.07	0.77
5	3.546	0.06	0.77	3.713	0.08	0.72	3.263	0.07	0.77
6	4.292	0.07	0.77	4.453	0.08	0.72	3.925	0.07	0.77
7	5.150	0.07	0.77	5.115	0.08	0.72	4.665	0.07	0.77
8	5.693	0.07	0.77	5.853	0.08	0.72	5.328	0.07	0.77
9	6.505	0.06	0.77	6.518	0.08	0.72	6.063	0.07	0.77
10	7.113	0.06	0.77	7.253	0.08	0.72	6.731	0.07	0.78

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11	7.912	0.06	0.80	7.92	0.08	0.72	7.464	0.07	0.77
12	8.534	0.06	0.81	8.655	0.08	0.71	8.133	0.07	0.76
13	9.315	0.06	0.78	9.321	0.08	0.71	8.867	0.07	0.76
14	9.938	0.06	0.77	10.057	0.08	0.72	9.533	0.07	0.76
15	10.716	0.06	0.77	10.724	0.08	0.73	10.269	0.07	0.76
16	11.34	0.06	0.77	11.459	0.08	0.73	10.934	0.07	0.75
17	12.118	0.06	0.77	12.125	0.08	0.73	11.670	0.07	0.76
18	12.741	0.06	0.77	12.861	0.08	0.73	12.337	0.07	0.75
19	13.520	0.07	0.77	13.525	0.08	0.72	13.070	0.08	0.75
20	14.143	0.07	0.76	14.264	0.08	0.71	13.739	0.08	0.75
21	14.923	0.07	0.76	14.926	0.08	0.72	14.472	0.08	0.75
22	15.544	0.07	0.76	15.666	0.08	0.72	15.140	0.08	0.74
23	16.325	0.07	0.76	16.327	0.08	0.71	15.874	0.08	0.74
24	16.946	0.07	0.77	17.068	0.09	0.71	16.540	0.08	0.74
25	17.726	0.07	0.76	17.728	0.09	0.71	17.276	0.08	0.74

Table 6:

15° Data Table (25 Data Points)									
No.	**Time (s) Run 1**	**Time in Gate (s) Run 1**	**Speed in Gate (m/s) Run 1**	**Time (s) Run 2**	**Time in Gate (s) Run 2**	**Speed in Gate (m/s) Run 2**	**Time (s) Run 3**	**Time in Gate (s) Run 3**	**Speed in Gate (m/s) Run 3**
1	0.414	0.04	1.27	0.424	0.04	1.30	0.438	0.04	1.26
2	1.058	0.04	1.29	1.061	0.04	1.27	0.963	0.04	1.26
3	1.796	0.04	1.26	1.806	0.04	1.23	1.712	0.04	1.2
4	2.436	0.04	1.25	2.441	0.04	1.22	2.342	0.04	1.19
5	3.181	0.04	1.19	3.183	0.04	1.23	3.091	0.04	1.17
6	3.811	0.04	1.22	3.821	0.04	1.21	3.721	0.04	1.17
7	4.561	0.04	1.22	4.562	0.04	1.21	4.471	0.04	1.14
8	5.190	0.04	1.21	5.199	0.04	1.20	5.1	0.04	1.15
9	5.940	0.04	1.20	5.944	0.04	1.20	5.851	0.04	1.15
10	6.570	0.04	1.19	6.576	0.04	1.2	6.479	0.04	1.14
11	7.318	0.04	1.20	7.323	0.04	1.21	7.227	0.04	1.16
12	7.952	0.04	1.18	7.956	0.04	1.21	7.86	0.04	1.14
13	8.698	0.04	1.19	8.699	0.04	1.19	8.606	0.05	1.13
14	9.332	0.04	1.16	9.336	0.04	1.18	9.237	0.04	1.13

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15	10.076	0.04	1.15	10.079	0.04	1.18	9.988	0.05	1.13
16	10.709	0.04	1.17	10.712	0.04	1.17	10.615	0.05	1.13
17	11.459	0.04	1.14	11.46	0.04	1.17	11.367	0.05	1.13
18	12.088	0.04	1.14	12.09	0.04	1.17	11.992	0.05	1.13
19	12.837	0.04	1.11	12.839	0.04	1.16	12.745	0.05	1.12
20	13.466	0.04	1.11	13.469	0.04	1.15	13.371	0.05	1.12
21	14.216	0.05	1.10	14.217	0.04	1.14	14.121	0.05	1.12
22	14.842	0.05	1.10	14.848	0.04	1.14	14.75	0.05	1.12
23	15.592	0.05	1.08	15.594	0.04	1.14	15.497	0.05	1.11
24	16.224	0.05	1.08	16.228	0.04	1.13	16.127	0.05	1.10
25	16.971	0.05	1.07	16.972	0.04	1.13	16.875	0.05	1.10

The two data tables presented above are excerpts from the 3° and 15° trials. When comparing the change in velocity of the data sets, it is apparent that the damping effect has increased with the angle.

-	**3°**	**15°**
Velocity at Data Point 1 (Average of Three Trials) (m/s)	0.777 ±0.001	1.277 ±0.001
Velocity at Data Point 25 (Average of Three Trials) (m/s)	0.737 ±0.001	1.100 ±0.001
Change in Velocity (m/s)	0.040 ±0.001	0.177 ±0.001

Sample Calculation of the Mean:

$$Velocity\ at\ Data\ Point\ 1\ = \frac{(0.8 + 0.75 + 0.78)}{3}$$

$$= 0.77667 \text{m/s}$$

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Uncertainty:

$$0.001 + 0.001 + 0.001 = 0.003$$

$$\frac{0.003}{3} = \pm 0.001s$$

Based on the calculations, the change in velocity for 15° is larger than that of 3° (0.17667 > 0.04). This demonstrates that the pendulum undergoes a greater damping effect when the release angle is larger, directly supporting the hypothesis. The same understanding can be derived by using the amplitude data.

2.2 Processed Data

To identify the damping coefficient and create a concrete conclusion to the experiment, it is necessary to process the raw data. For the Time and Amplitude, the mean and standard deviation of all 5 trials are presented below. This calculation took the final data values from the entire data set. All data points extracted from the SPARKvue software have an uncertainty of ±0.001.

Uncertainty of the Mean:

$$0.001 + 0.001 + 0.001 + 0.001 + 0.001 = 0.005$$

$$\frac{0.005}{5} = \pm 0.001s$$

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Uncertainty of the Standard Deviation:

$$\Delta\sigma = \frac{uncertainty\ of\ each\ data\ point}{\sigma \cdot \sqrt{2n}}$$

Where:

σ - Standard deviation (0.6969)

n - Number of data points (5)

Uncertainty for each data point - ±0.001

$$\sigma \cdot \sqrt{2n} = 0.6969 \cdot \sqrt{10}$$

$$= 2.203$$

$$\Delta\sigma = \frac{0.001}{2.203}$$

$$= \pm 0.000454$$

$$\approx \pm 0.0005s$$

Calculations for the Time:

Angle (± 0.05) (deg)	Mean (x̄) (±0.001) (s)	Standard Deviation (σ)
3.00	161.291	0.697
6.00	165.776	0.433
9.00	168.877	0.852
12.00	171.678	0.364
15.00	175.157	0.663

Sample Calculation of the Mean:

$$\text{Mean Time of } 3° = \frac{(162.3 + 160.78 + 160.317 + 161.318 + 161.738)}{5}$$

$$= 161.291s$$

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Sample Calculation of the Standard Deviation:

1) Subtracting the Data Values from the Mean	2) Squaring each Deviation	3) Finding the Variance
$162.3-161.2906 = 1.0094$ $160.78-161.2906 = -0.5106$ $160.317-161.2906 = -0.9736$ $161.318-161.2906 = 0.0274$ $161.738-161.2906 = 0.4474$	$(1.0094)^2 = 1.0189$ $(-0.5106)^2 = 0.2607$ $(-0.9736)^2 = 0.9479$ $(0.0274)^2 = 0.0008$ $(0.4474)^2 = 0.2002$	$\frac{(1.0189+0.2607+0.9479+0.0008+0.2002)}{5}$ $= 0.4857$
4) Square root of the Variance (Standard Deviation):		$\sqrt{0.4857} = 0.6969$

Calculations for the Amplitude:

Angle (± 0.05) (deg)	**Mean ($\bar{x}$) (±0.001)**	**Standard Deviation (σ)**
3.00	0.0511	0.00174
6.00	0.0593	0.00202
9.00	0.0624	0.00172
12.00	0.0682	0.00279
15.00	0.0753	0.00199

Sample Calculation of the Mean:

$$\text{Mean Amplitude of } 3° = \frac{(0.05 + 0.0498 + 0.05134 + 0.05437 + 0.04989)}{5}$$

$$= 0.0511\text{m}$$

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Sample Calculation of the Standard Deviation:

1) Subtracting the Data Values from the Mean	2) Squaring each Deviation	3) Finding the Variance
0.05−0.05108 = −0.00108 0.0498−0.05108 = −0.00128 0.05134−0.05108 = 0.00026 0.05437−0.05108 = 0.00329 0.04989−0.05108 = −0.00119	$(-0.00108)^2 = 0.0000011664$ $(-0.00128)^2 = 0.0000016384$ $(0.00026)^2 = 0.0000000676$ $(0.00329)^2 = 0.0000108241$ $(-0.00119)^2 = 0.0000014161$	[(0.000001164 + 0.0000016384+0.0000000676+0.000010824+0.0000014161)]/5 = 0.0000030225
4) Square root of the Variance (Standard Deviation):		√0.0000030225 = 0.001738

By calculating the standard deviation, it is evident that the data for the Time and Amplitude is precise.

This adds to the reliability of the data set as it has a low spread.

Table 7:

Combined Mean for Time (s) and Amplitude (m)		
Angle (± 0.05) (deg)	**Mean for Time (s) ($\bar{x}$) (±0.001)**	**Mean for Amplitude (m) ($\bar{x}$) (±0.001)**
3.00	161.291	0.051
6.00	165.776	0.059
9.00	168.877	0.062
12.00	171.678	0.068
15.00	175.157	0.075

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As the mean of each data set has been calculated, we can utilise the derived formula to identify the damping coefficient.

Damping Coefficient:

Table 8:

Damping Coefficient	
Angle (± 0.05) (deg)	**Damping Coefficient (Ns/m) (±0.0021)**
3.00	0.0003
6.00	0.0068
9.00	0.0108
12.00	0.0129
15.00	0.0141

Sample Calculation:

$$k = -2\frac{ln(\frac{\theta}{Lsin(\theta_1)})}{t}$$

$$k = -2\frac{ln(\frac{0.05108}{1sin(0.0523599)})}{161.291}$$

$$= 0.00030127528$$

$$\approx 0.0003 \text{Ns/m}$$

Uncertainty Calculation:

The calculation for the uncertainty of this formula is complex, hence, a shortened version will be presented. The formula below has been derived using principles of uncertainty, particularly with the trigonometric rules. By substituting the values of each uncertainty into the formula, the final uncertainty for **k is ± 0.0021.**

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$$\Delta k = \sqrt{\left(\frac{2}{t \cdot \theta}\Delta\theta\right)^2 + \left(\frac{2}{t \cdot L}\ln\left(\frac{\theta}{L\sin(\theta_1)}\right)\Delta L\right)^2 + \left(\frac{2}{t \cdot \sin(\theta_1)}\ln\left(\frac{\theta}{L\sin(\theta_1)}\right)|\cos(\theta_1)|\Delta\theta_1\right)^2 + \left(\frac{2}{t^2}\ln\left(\frac{\theta}{L\sin(\theta_1)}\right)\Delta t\right)^2}$$

Where:

θ_1 (angle in radians) - ±0.05

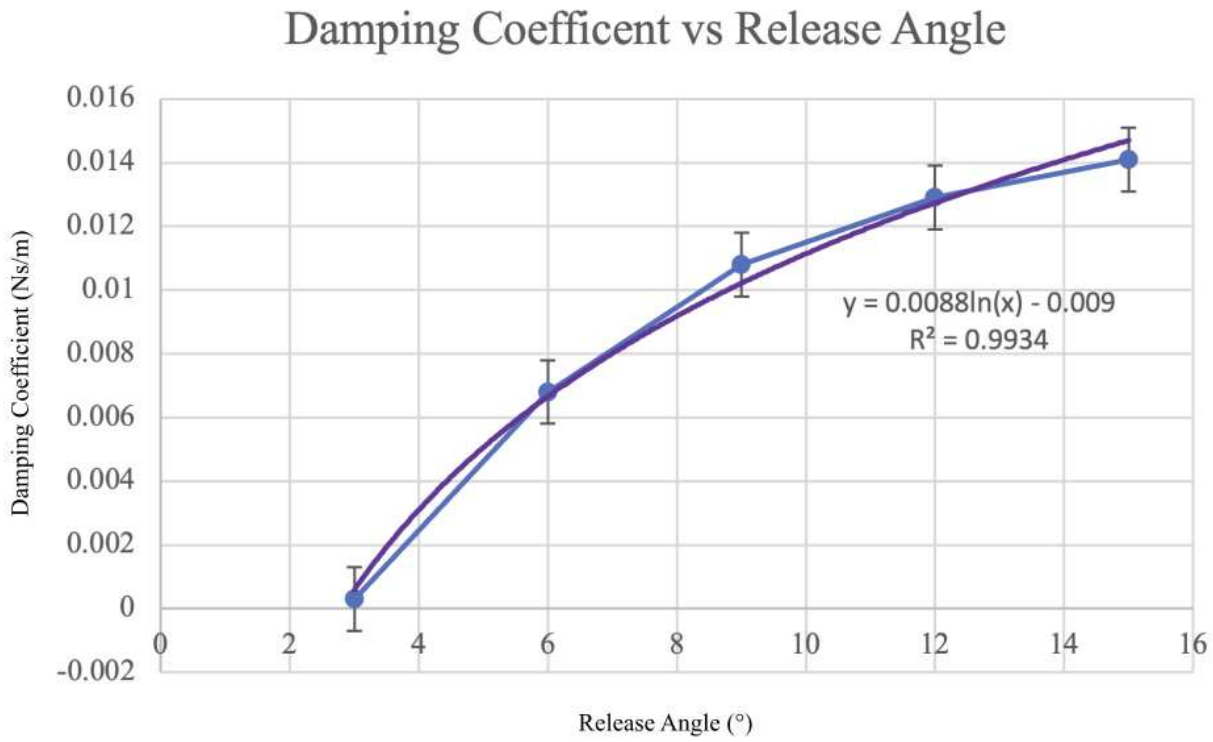
θ (amplitude at 2 seconds) - ±0.001

L (length of the string in meters) - ±0.005

t (time in seconds) - ±0.001

2.3 Results - Graphical Review

By using the derived formula, the data for the Damping Coefficient corresponds to the original hypothesis and the information collected during qualitative analysis. To visualise this relationship, a model is presented below with error bars and accurate data points.

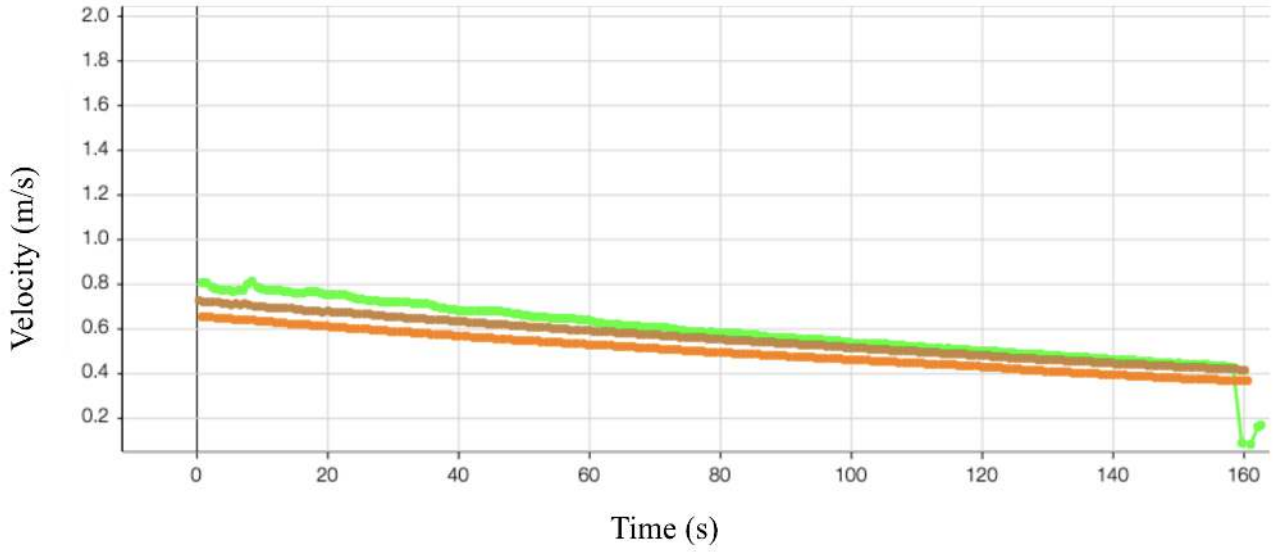


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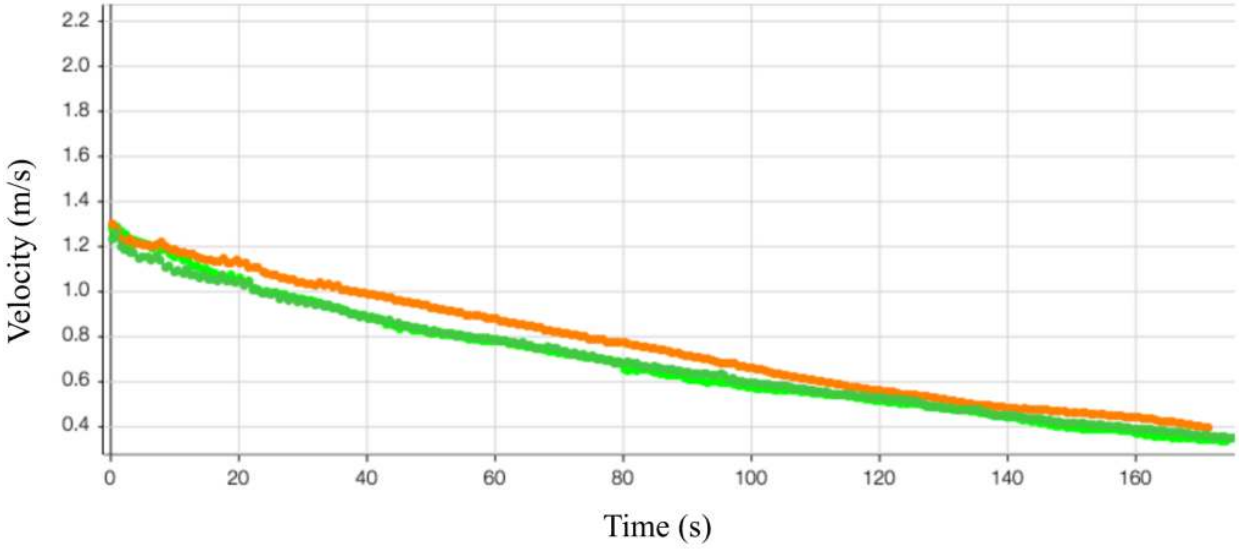
As the release angle increases from 3° to 15°, the damping coefficient shows a consistent upward trend, indicating a proportional relationship between the two variables. This is further supported by a strong correlation, as reflected by a Pearson correlation coefficient of 0.9934.

After observing graphs from the SPARKvue software, this relationship is proven even further. Below is a comparison of the 3° to 15° graphs.

The 3° Graph:



The 15° Graph:



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The variation in the slopes of the graphs demonstrates the damping effect in action. As the release angle increases, the damping effect intensifies, leading to a steeper curve. This signifies a more rapid loss of velocity.

3. Conclusion

Overall, the research question “To what extent does an increase in the release angle of a pendulum (3°, 6°, 9°, 12°, 15°) influence its damping coefficient (Ns/m) in Simple Harmonic Motion when keeping the mass of the ball and length of the string constant? ” posed an intriguing investigation and line of inquiry. The relationship between the release angle and the damping coefficient became more apparent through the process, especially with a Pearson Correlation Coefficient of 0.9934. This finding underscores a fascinating dynamic in the behaviour of pendulums where an increase in the release angle is met with a larger damping coefficient.

The experiment was heavily based on a formula derived from the Mohazabbi and Shankar study. While their research primarily examined the damping coefficient, the pendulum bob mass, and the string's diameter, their findings were highly relevant to this investigation. Their work established clear relationships between damping, time, and amplitude, providing a solid scientific foundation. This ensured that the study remained focused and grounded in well-established principles. Moreover, by building on these principles, the experiment created grounds for the evolution of knowledge as existing prerequisites were used to identify a new relationship between the physical space and the movement of pendulums.

As a whole, this led to a reliable and scientific process where ideas were formed and proven utilising the empirical method. This highlighted the nature of exploration and the power of discovery for knowledge creation.

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4. Evaluation

4.1 Uncertainty:

As a large part of the data was collected using digital equipment, the margin for error is minimal and often negligible. However, to account for potential systematic errors, uncertainties were assigned to the data from the SPARKvue software, such as time and amplitude, with a margin of ±0.001. Additionally, to address possible human errors in manually recorded data, uncertainties for variables such as mass, length, and release angle were recorded. The derived formula also underwent uncertainty analysis, resulting in a calculated margin of ±0.002 for the damping coefficient. This consideration of uncertainties ensured the reliability of the data and results.

4.2 Weaknesses and Limitations

Random Errors		
Error Source	**Impact (Explanation of Error)**	**Improvement**
Uncertainty with the Angle Measurement	As the SPARKvue software and Pasco smart gate did not have an inbuilt release angle system, a protractor was used instead. While the angle was measured many times to ensure reliability, it paved the way for human error.	Instead of relying on external equipment to measure the release angle, software with the inbuilt data should be used.
Unexpected Environmental Disturbances	The experimentation room had a slight breeze during the process, which could have influenced the results. While the number of trials limits this problem, it still creates a natural disturbance.	A more controlled environment should be used with proper equipment to limit any external factors.
Release of the Pendulum	As the pendulum was released by hand, a different force was applied with each release. While the difference was minimal, it could have affected the results.	To use a projectile launcher or launch stand where human error is completely minimised.

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Systematic Errors		
Error Source	**Impact (Explanation of Error)**	**Improvement**
Calibration of the Pasco Smart Gate	The Smart Gate was a recently purchased piece of equipment. This could have resulted in calibration issues, especially when setting up the different modes and connecting it to the laptop.	Test out the different modes to identify the best one for the experimentation process.
Calibration of the SPARKvue Software	Along with the Smart Gate, the SPARKvue software was new and previously untested. This left calibration up to chance and created room for systematic error.	To conduct numerous trials with the software, systematic error is limited as the software is effectively calibrated.
Sample Size	As the sample size included only 5 angles, the effect of systematic errors on the data was much higher. This has the possibility of skewing the results or creating outliers.	Include a large sample size with different angles below 15° to limit the effect of errors.

4.3 Strengths

Strength	**Evaluation**
Use of Digital Equipment	As the experimentation process relied on digital equipment to record and process the data, each data point was reliable and precise. This ensured that all results were credible and effectively supported the hypothesis.
Uncertainty Analysis	The procedure acknowledged the room for uncertainties and assigned values to each variable. This created flexibility in the final range of values while recognising the margin for error.
Solid Theoretical Foundation	The entire experiment was based on the procedure by Mohazabbi and Shankar. This created a sound theoretical foundation to build upon as all pieces of information came from a reliable source.
Novel Insight	While references were taken, this experiment investigates a novel variable. This creates the opportunity to expand on knowledge and offer new perspectives in the scientific community.
Focused and Structured approach	The entire procedure is easily replicable due to its focus and structure. This limits any unnecessary complexities while ensuring that other learners can evaluate the results.

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4.4 Extension of the investigation

The pendulum is an object that exists across the world. The possibilities for its use, just like the possibility for experimentation, are practically endless. By considering other variables such as the shape of the mass, the pattern of revolution, or even the diameter of the wire, learners can develop their understanding of the world. Experiments can even connect to broader areas of study that are universally applicable. Seismology is one such example as pendulums help create predictions and allow individuals to better understand the world beneath the surface. By continuing the path of inquiry, new sets of knowledge will emerge. This will better the world by opening the door to possibilities that were previously inaccessible.

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